Sparse-Sample Reconstruction of Crop Vigor Using Sentinel-2 NDVI: A Preliminary Benchmark of IDW and Kriging for Precision Agriculture

Mahbubul Alam[1,*] · Abdullah All Mamun[2]

[1] Department of Forest Information Technology, Eberswalde University for Sustainable Development (HNEE), 16225 Eberswalde, Germany
[2] Information and Communication Engineering, Berliner Hochschule für Technik (BHT), 13353 Berlin, Germany
* **Correspondence:** u23112@hnee.de · **Co-author:** hvnn3504@bht-berlin.de

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At field-sampling densities, classical geostatistics remains the most reliable NDVI reconstructor: no AI predictor surpassed it from 20 to 200 samples.

- **0.092** Full-field RMSE of ordinary kriging, best of 7 methods
- **p = 0.38** IDW vs. kriging: statistically indistinguishable
- **+6–10%** RMSE penalty of the best ML/DL predictor
- **0.88** Kriging RMSSE: calibrated uncertainty (ideal 1)
- **81.4%** Management-zone agreement with reference

1 Introduction

Cloud cover and sparse field measurements leave gaps in the NDVI maps that precision agriculture relies on.

Can AI predictors reconstruct field-scale NDVI from sparse samples more accurately than classical geostatistics?

- Benchmark 2 classical and 5 learning-based predictors under identical, paired conditions.
- Score held-out points **and** the whole parcel over 50 repeated draws.
- Track the gap across sampling densities.

2 Study Area & Data

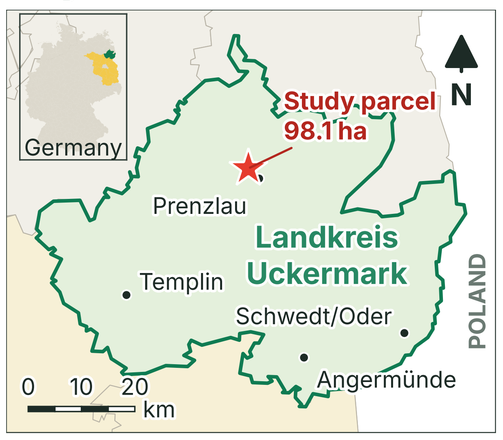


Fig. 1. Study parcel near Prenzlau, Landkreis Uckermark, Brandenburg (ETRS89 / UTM 33N).

Imagery	**Sentinel-2 L2A, 7 scenes, Jun–Jul 2024**
Parcel	**98.1 ha · 9,811 pixels at 10 m**
NDVI	**0.757 ± 0.144 (range 0.24 – 0.90)**

3 Methodology

1 Simulate sparse sampling
n = 50 points from the reference raster + Gaussian noise (σ = 0.03); 80 / 20 split; 50 repetitions.

2 Fit 7 predictors on identical points
Hyperparameters tuned inside each run on training data only.

3 Validate twice
10 held-out points and all 9,811 pixels per run (RMSE, MAE, R^2).

4 Test significance
Friedman; Wilcoxon + Holm; paired t-test (IDW vs. OK).

5 Extend
Density sweep n = 20 – 200; RMSSE; 3-class k-means zones.

Seven predictors, four families

CLASSICAL
IDW (LOOCV-tuned) · Ordinary kriging (OK)
AI-BASED
Random forest (RF) · Gradient boosting (GB) · Gaussian process (GP) · Fourier-feature MLP · RF + residual kriging

Learners use K-nearest-sample spatial features without leakage. OK uses an **exponential variogram** (range 1,129 m), best of four by LOOCV.

4 Results

4.1 Accuracy ranking n = 50 · 50 paired runs

Method	Full field	Held-out	R^2
Ordinary kriging		**0.089**	**0.59**
IDW (LOOCV-tuned)		**0.090**	**0.56**
RF + residual kriging		0.096	0.53
Gaussian process		0.097	0.53
Fourier-feature MLP		0.103	0.49
Random forest		0.098	0.50
Gradient boosting		0.110	0.38

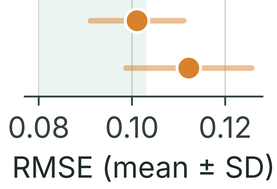


Fig. 2. Full-field RMSE mean ± SD; held-out RMSE and full-field R^2 as means.

Methods differ
Friedman χ^2 = 35.6, p < 0.001

IDW ≈ kriging
Δ RMSE 0.001, p = 0.38

Classical > AI
vs. all 5 learners, p ≤ 0.021

Single splits mislead
R^2 spans −9.2 to 0.97

4.2 Effect of sampling density 30 runs per n

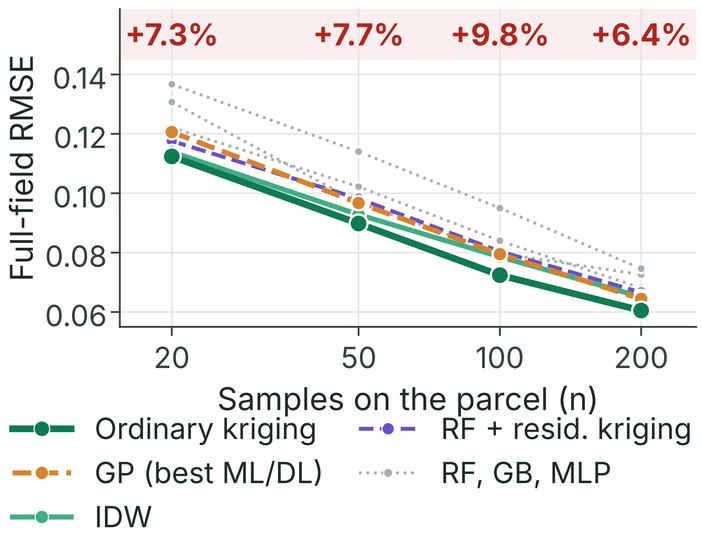


Fig. 3. Red: RMSE penalty of the best ML/DL predictor (GP) vs. kriging. No crossover up to 2 samples ha^{-1}.

4.3 Reconstructions same 50 samples

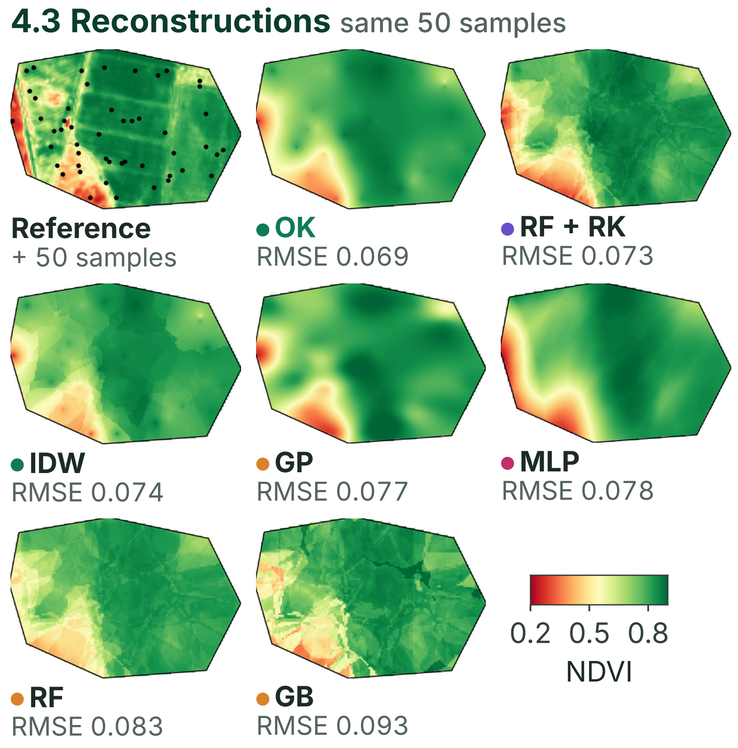
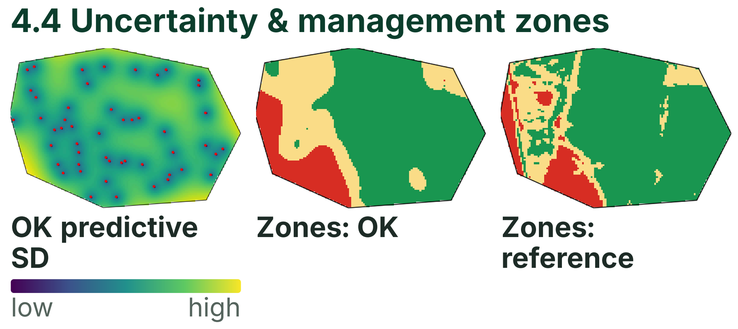


Fig. 4. Single run, ordered by RMSE. OK keeps the low-vigor patches; learners smooth (GP, MLP) or fragment (RF, GB) them.

4.4 Uncertainty & management zones



OK uncertainty is calibrated (RMSSE 0.88; GP 1.12) and flags where to sample next. Zone agreement: GP 82.0 %, RF 81.9 %, **OK 81.4 %**, IDW 72.3 %.

5 Conclusions

- At 0.2 – 2 samples ha^{-1}, **kriging and IDW** are the most reliable and statistically equivalent.
- No AI predictor surpassed geostatistics at any tested density: a **data-regime limit**, not an architectural failure.
- Kriging adds calibrated per-pixel uncertainty for adaptive sampling.
- For AI in remote sensing, the benchmark marks the data threshold below which learned predictors should not replace geostatistics.

Limitations & outlook
One parcel and season; simulated samples; zones not validated against yield. Next: multi-parcel, multi-season validation and transfer learning.

Abbreviations **AI** artificial intelligence · **CI** confidence interval · **DL** deep learning · **ETRS89** European Terrestrial Reference System 1989 · **GADM** Database of Global Administrative Areas · **GB** gradient boosting · **GP** Gaussian process (regression) · **IDW** inverse distance weighting · **k, K** number of nearest neighbouring samples · **L2A** Level-2A surface reflectance · **LOOCV** leave-one-out cross-validation · **MAE** mean absolute error · **ML** machine learning · **MLP** multilayer perceptron (neural network) · **n** number of samples · **NDVI** Normalized Difference Vegetation Index · **OK** ordinary kriging · **p** p-value (significance) · **R^2** coefficient of determination · **RF** random forest · **RF + RK** random forest + residual kriging · **RMSE** root-mean-square error · **RMSSE** root-mean-square standardised error · **SD** standard deviation · **UTM** Universal Transverse Mercator · **σ** SD of simulated sensor noise · **χ^2** Friedman test statistic

References [1] Oliver & Webster, *Catena* 2014 · [2] Hengl et al., *PeerJ* 2018 · [3] Tancik et al., *NeurIPS* 2020. **Data** modified Copernicus Sentinel-2 (2024) via Google Earth Engine; GADM; PyKrige, scikit-learn, rasterio.

Full project, code & data
Scan to open the GitHub repository with the complete notebook and results.
github.com/allmamun556/Sparse-Sample-Reconstruction-of-Crop-Vigor-Using-Sentinel-2-NDVI